

MTH:3340 Abstract Algebra Exam 1 Reassessment

Complete all of the following problems being as detailed as you can be. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit. You may not use a calculator, notes, or other resources.

Handwriting: Your ability to express your solutions is just as important as mathematical accuracy.

You are expected to write every solution using complete sentences when possible.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

1. (14 points) Show that the set $S = \{x \mid x = a + b\sqrt{2} \text{ where } a, b \in \mathbb{Q} \text{ and } a \neq 0\}$ is a group with respect to addition.

Let $x, y \in S$ then $x = a + b\sqrt{2}$, $y = c + d\sqrt{2}$ for $a, b, c, d \in \mathbb{Z}$

so $x + y = (a + c) + (b + d)\sqrt{2} \in S$ and always defined

so $+$ is a binary operation on S .

Identity: $0 + 0\sqrt{2} = 0 \in S$ and

$0 + (x) = 0 + (a + b\sqrt{2}) = a + b\sqrt{2}$ for any $x \in S$.

Inverses: $x + y = 0$ if $(a + c) + (b + d)\sqrt{2} = 0$ with

unique solution $\bar{x} = -a - b\sqrt{2} \in S$ so inverses exist.

Assoc: $x + (y + z)$ is addition based, so assoc.

$$\begin{aligned} x + (y + z) &= a + b\sqrt{2} + (c + d\sqrt{2} + e + f\sqrt{2}) \\ &= (a + c + e) + (b + d + f)\sqrt{2} \\ &= (a + b\sqrt{2} + c + d\sqrt{2}) + (e + f\sqrt{2}) \\ &= (x + y) + z \quad \checkmark \end{aligned}$$

2. (8 points) Let $(G, \circ)$ be a group with identity e . Prove that if $g^2 = e$ for all elements $g \in G$, then G is abelian.

Let $(G, \circ)$ be a group with identity e .

Assume $g^2 = e$ for all $g \in G$.

For $g, h \in G$, then $(gh)^2 = e$, $g^2 = e$, $h^2 = e$

so $g^{-1} = g$ and $h^{-1} = h$. So

$$(gh)^2 = e \Rightarrow ghgh = e$$

$$\Rightarrow hgh = g \Rightarrow hg = gh$$

so G is abelian.

Therefore, if $g^2 = e$ for all elements g in a group G , G is abelian. ~~QED~~